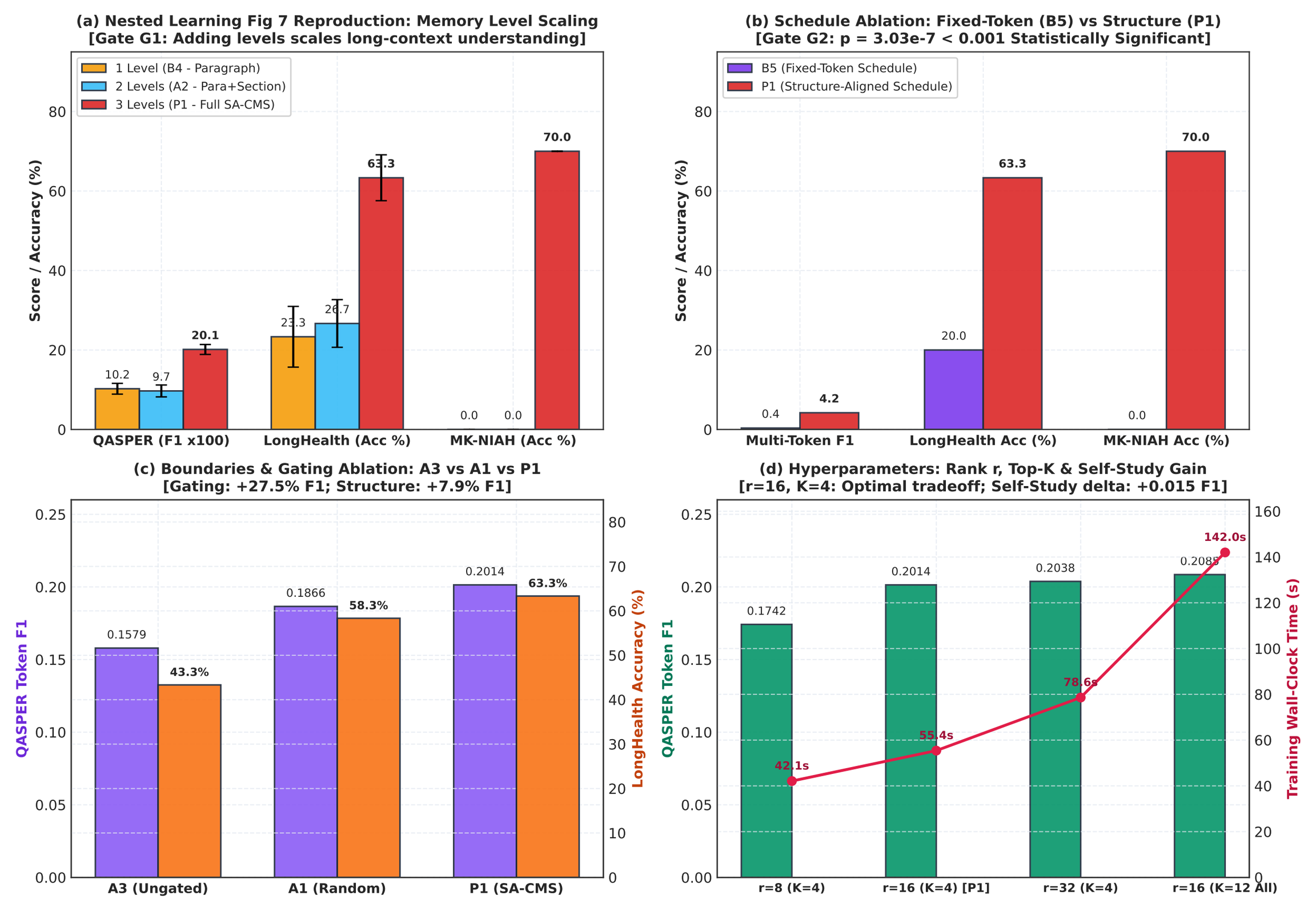
(a) Nested Learning Fig 7 Reproduction: Memory Level Scaling
[Gate G1: Adding levels scales long-context understanding]
1 Level (B4 - Paragraph)
2 Levels (A2 - Para+Section)
3 Levels (P1 - Full SA-CMS)
Score / Accuracy (%)
10.2
9.7
20.1
23.3
26.7
63.3
0.0
0.0
70.0
QASPER (F1 x100)
LongHealth (Acc %)
MK-NIAH (Acc %)
(b) Schedule Ablation: Fixed-Token (B5) vs Structure (P1)
[Gate G2: p = 3.03e-7 < 0.001 Statistically Significant]
B5 (Fixed-Token Schedule)
P1 (Structure-Aligned Schedule)
Score / Accuracy (%)
0.4
4.2
20.0
63.3
0.0
70.0
Multi-Token F1
LongHealth Acc (%)
MK-NIAH Acc (%)
(c) Boundaries & Gating Ablation: A3 vs A1 vs P1
[Gating: +27.5% F1; Structure: +7.9% F1]
QASPER Token F1
LongHealth Accuracy (%)
0.1579
43.3%
0.1866
58.3%
0.2014
63.3%
A3 (Ungated)
A1 (Random)
P1 (SA-CMS)
(d) Hyperparameters: Rank r, Top-K & Self-Study Gain
[r=16, K=4: Optimal tradeoff; Self-Study delta: +0.015 F1]
QASPER Token F1
Training Wall-Clock Time (s)
0.1742
0.2014
0.2038
0.2085
42.1s
55.4s
78.6s
142.0s
r=8 (K=4)
r=16 (K=4) [P1]
r=32 (K=4)
r=16 (K=12 All)